$Re_{inj} = 100$ $Re_{inj} = 200$ $Re_{inj} = 300$ Fine grid + half step (Re = 300) Second seed (Re = 300)

a Final 3D kinetic-energy fraction

$100\chi_E(1200)$ (%)

Initial $u'_{3D,0}/u_{s0}$

b Absolute 3D energy: growth or decay

$E_\perp(1200)/E_\perp(0)$

growth > 1
decay < 1

Initial $u'_{3D,0}/u_{s0}$

c Departure of the mean current

$100\varepsilon_U(1200)$ (%)

Initial $u'_{3D,0}/u_{s0}$

d Background shear changes 3D motion

$100\chi_E(1200)$ (%)

$\eta = 10^{-6}$
$\eta = 1$

Background parameter α $(q_0 = -\alpha u'_s)$

e Background shear changes mean feedback

$100\varepsilon_U(1200)$ (%)

$\eta = 10^{-6}$
$\eta = 1$

Background parameter α $(q_0 = -\alpha u'_s)$

f Effect of waves on 3D energy

Waves on
Waves off

$E_\perp(1200)/E_\perp(0)$

$\alpha = 0.75,\ \eta = 10^{-6}$

Injection-scale $Re_{inj} = U_{2D}\ell/\nu$